# CSCI 164 Report - Hazardous Asteroid Classification with scikit-learn

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## Summary

This project applies supervised machine learning to two related asteroid datasets using scikit-learn. The first dataset is the Asteroid Dataset, where the target variable is **pha**, meaning Potentially Hazardous Asteroid. The second dataset is the NeoWs Hazardous Asteroid Dataset, where the target variable is **Hazardous**, meaning whether a close-approach record is hazardous or not.

The models used were Logistic Regression, K-Nearest Neighbors, and Multi-Layer Perceptron. Each model was trained on both datasets, tuned with GridSearchCV, and evaluated with accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. The best model for the asteroid dataset was Logistic Regression, while the best model for the close-approach dataset was MLP.

The main finding is that the close-approach dataset was easier for the models to classify. The asteroid dataset was more difficult because the hazardous class was rare, meaning that some models had high accuracy while failing to identify hazardous records. This supports the same general issue discussed in the referenced paper: hazardous asteroid classification is possible with machine learning, but class imbalance strongly affects model performance.

## 1.1) Dataset Selection

Source - https://arxiv.org/pdf/2409.02150

Google Colab - https://colab.research.google.com/drive/1CJoe9fkw6b-fRoTy4u3Ssu8n_OBIzasu?usp=sharing

GitHub - https://github.com/Dexless/CSCI-164-Colab-Repo/blob/main/164Project.ipynb

Asteroid Dataset (https://www.kaggle.com/datasets/sakhawat18/asteroid-dataset) - The Asteroid Dataset is a dataset compiled from NASA/JPL asteroid records. The original dataset contains 958,524 rows and 45 total features. It includes asteroid identifiers, orbital measurements, magnitude values, orbit classifications, and hazardous asteroid labels. In this project the target attribute is **pha**, which identifies whether an asteroid is a Potentially Hazardous Asteroid.

NeoWs Dataset (https://www.kaggle.com/datasets/alvinb/neows-hazardous-asteroid-dataset) - The NeoWs Hazardous Asteroid Dataset contains near-Earth object close-approach records. It includes estimated diameter, velocity, miss distance, close-approach dates, orbital features, and hazardous classification. In this project the target attribute is **Hazardous**, which identifies whether a close-approach record is hazardous.

These datasets were selected because they are related by domain but support two separate supervised learning tasks. The asteroid dataset focuses on asteroid-level properties, while the close-approach dataset focuses on event-level properties. The datasets were not merged because the project requires multiple datasets, and keeping them separate makes the model comparison clearer.

## 1.2) Features, Targets, and Data Types

| Dataset | Target | Problem Type |
|---|---|---|
| Asteroid Dataset | `pha` | Binary classification |
| NeoWs Dataset | `Hazardous` | Binary classification |

Several important terms are used throughout the project. **PHA** means Potentially Hazardous Asteroid. **NEO** means Near-Earth Object. **H** or absolute magnitude is a brightness measurement where lower values often indicate brighter or larger objects. **MOID** means Minimum Orbit Intersection Distance, which measures how

close an asteroid's orbit comes to Earth's orbit. **Orbit class** describes the asteroid's orbital group, such as APO, ATE, AMO, IEO, or MCA.

These definitions are included because the model results depend on domain-specific attributes like MOID, orbit class, velocity, miss distance, and hazardous status.

## 1.3) Prior Work

The referenced paper is *Hazardous Asteroids Classification*. The paper uses asteroid and NeoWs-related datasets to classify hazardous asteroids with machine learning and deep learning methods. It also discusses class imbalance and mentions balancing methods such as SMOTE and bootstrapping.

This project follows the same general topic, but it is narrower in scope. Instead of using many deep learning models, this project focuses on scikit-learn models that match the course requirements. SMOTE was not used in this project. Instead, stratified splitting was used to preserve class proportions, and Logistic Regression used balanced class weights to help account for the rare hazardous class.

## 2.1) Preprocessing

The preprocessing stage removed unusable attributes, handled missing data, encoded categorical values, and scaled numeric features.

For the asteroid dataset, columns with very high missing percentages were removed. These included:

- `prefix`
- `name`
- `albedo`
- `diameter`
- `diameter_sigma`

These columns were missing roughly 85% to almost 100% of their values, so keeping them would have added too much missing data without clearly improving the model. Rows missing the target value **pha** were also removed because supervised learning requires known labels.

For the close-approach dataset, redundant measurement columns were removed. Several columns represented the same idea in different units. Keeping only one version of these measurements helped reduce repeated information.

The asteroid dataset also required categorical transformation. The **neo** and **pha** values were converted from Y/N values into numeric values. The orbit class column was one-hot encoded because scikit-learn models require numeric input. For the NeoWs dataset, close-approach dates were converted into year, month, day, and dayofyear features.

Both datasets were sampled to 30,000 records to keep the notebook manageable in Google Colab. The data was split into 80% training, 10% validation, and 10% test. Stratification was used so that each split kept roughly the same target class distribution. Median imputation handled missing numeric feature values, and StandardScaler was used because Logistic Regression, KNN, and MLP are affected by feature scale.

## 2.2) Model Development

Three supervised learning algorithms were used for each dataset: Logistic Regression, K-Nearest Neighbors, Multi-Layer Perceptron

These models were chosen because they are available in scikit-learn and are appropriate for binary classification. Logistic Regression was especially useful because the hazardous class is rare. KNN gave a different type of model comparison, and MLP provided a more flexible nonlinear method.

Each model was first trained as a baseline model and evaluated on the validation set. The metrics used were accuracy, precision, recall, F1-score, and ROC-AUC.

## 3.1) Baseline Asteroid Dataset Results

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.990333 | 0.194444 | 1.000000 | 0.325581 | 0.996277 |
| KNN | 0.997000 | 0.000000 | 0.000000 | 0.000000 | 0.783280 |
| MLP | 0.997000 | 0.000000 | 0.000000 | 0.000000 | 0.994988 |

The baseline asteroid results show why accuracy is not enough for this dataset. KNN and MLP both had high accuracy, but they had 0 recall and 0 F1-score for the hazardous class. This means they mostly predicted the majority class and failed to identify hazardous asteroid records.

Logistic Regression had lower accuracy, but it had perfect recall on the validation set. This made it more useful for the asteroid dataset because missing a hazardous asteroid would be worse than flagging extra false positives.

## 3.2) Baseline Close-Approach Dataset Results

| Model | Accuracy | Precission | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.937667 | 0.687616 | 0.953846 | 0.799141 | 0.989513 |
| KNN | 0.907333 | 0.701439 | 0.500000 | 0.583832 | 0.917464 |
| MLP | 0.992000 | 0.966837 | 0.971795 | 0.969309 | 0.999519 |

The close-approach dataset produced stronger baseline results. MLP performed best overall, with the highest accuracy, precision, recall, F1-score, and ROC-AUC. Logistic Regression also performed well, especially for recall. KNN was weaker because its recall and F1-score were lower than the other models.

## 4.1) Hyperparameter Tuning

GridSearchCV was used to tune each model. F1-score was the main metric used because it balances precision and recall, which is better than accuracy for imbalanced classification problems.

| Model | Best Parameters (Asteroid) | Best Parameters (Close Approach) |
|---|---|---|
| Logistic Regression | `C = 10` | `C = 10` |
| KNN | `n_neighbors = 3`, `weights = uniform` | `n_neighbors = 3`, `weights = distance` |
| MLP | `alpha = 0.0001`, `hidden_layer_sizes = (50,)` | `alpha = 0.0001`, `hidden_layer_sizes = (25,)` |

The tuning results showed that model performance depended heavily on the dataset. Logistic Regression was strongest for the asteroid dataset, while MLP was strongest for the close-approach dataset.

## 4.2) Tuned Asteroid Dataset Results

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.992333 | 0.233333 | 1.000000 | 0.378378 | 0.997948 |
| KNN | 0.996667 | 0.000000 | 0.000000 | 0.000000 | 0.783590 |
| MLP | 0.997333 | 0.000000 | 0.000000 | 0.000000 | 0.350294 |

After tuning, Logistic Regression remained the best asteroid model. Its F1-score improved from 0.325581 to 0.378378, and it kept perfect recall. KNN and MLP still failed to identify the hazardous class, even though their accuracy was high.

This result shows that the asteroid dataset is strongly affected by class imbalance.

## 4.3) Tuned Close-Approach Dataset Results

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.938000 | 0.688889 | 0.953846 | 0.800000 | 0.989484 |
| KNN | 0.908667 | 0.680124 | 0.561538 | 0.615169 | 0.889105 |
| MLP | 0.991333 | 0.955000 | 0.979487 | 0.967089 | 0.999179 |

After tuning, MLP remained the best close-approach model. It had the strongest F1-score and roc-auc, along with a strong balance between precision and recall. Logistic Regression was still strong for recall but had lower precision. KNN improved slightly but remained the weakest model.

## 5.1) Final Model Comparison

| Dataset | Best Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|---|
| Asteroid | Logistic Regression | 0.989667 | 0.162162 | 1.000000 | 0.279070 | 0.999388 |
| Close Approach | MLP | 0.991667 | 0.971576 | 0.964103 | 0.967825 | 0.999330 |

The best models were selected using validation-set F1-score after GridSearchCV tuning. After the best model was selected for each dataset, the held-out test set was used for final evaluation.

## 5.2) Visual Results

The confusion matrices and metric graphs were included to show model behavior more clearly than tables alone.

For the asteroid dataset, the confusion matrix shows that Logistic Regression found the hazardous validation records but also produced false positives. For the close-approach dataset, the MLP confusion matrix showed stronger overall classification with fewer errors.

The metric graphs also support the same finding. The asteroid dataset was harder to evaluate with accuracy alone, while the close-approach dataset showed a clearer advantage for MLP.

## 6.1) Comparison to Prior Work

The referenced paper also treats hazardous asteroid prediction as a classification problem. It uses multiple machine learning and deep learning models, while this project focuses only on scikit-learn models required for class.

The paper discusses class balancing methods such as SMOTE and bootstrapping. This project did not use SMOTE, which could explain why KNN and MLP struggled on the asteroid datasets. Instead, this project used stratified splitting and balanced class weights for Logistic Regression.

The results support the same conclusion as the paper. Machine learning *can* classify hazardous asteroid records, but the results depend heavily on the dataset and on how class imbalance is handled.

## 6.2) Strengths and Limitations

A strength of this project is that it uses two related datasets and compares the same models across both of them. Another strength is that the evaluation uses several metrics instead of only accuracy.

A limitation is that the asteroid dataset is highly imbalanced. The hazardous class is very small, which makes precision and F1-score difficult to improve. Another limitation is that only 30,000 records were sampled from each dataset to keep Google Colab runtime manageable. Finally, SMOTE was not used, so in the future, testing the use of SMOTE could improves performance.

## 7.1) Conclusion

This project used two asteroid-related datasets to train and compare supervised classification models with scikit-learn. The models used were Logistic Regression, KNN, and MLPClassifier. Each model was trained as a

baseline model, tuned with GridSearchCV, and evaluated with accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and metric visualizations.

The best model for the asteroid dataset was Logistic Regression. It had low precision but perfect recall, making it the most useful model for identifying the rare hazardous class. The best model for the close-approach dataset was MLP, which performed strongly across nearly all metrics.

Overall, the close-approach dataset was easier to classify, while the asteroid dataset showed how class imbalance can hide poor model performance behind high accuracy. This project demonstrates that machine learning can be useful for hazardous asteroid classification, but evaluation must go beyond accuracy when the target class is rare.